LED Strobe Light Bell

A Classroom Visual Alert System

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For: Design for Independence

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Purpose and Goal of the LED Strobe Light Bell

The purpose of the LED Strobe Light Bell is to provide a reliable visual alert for students and teachers at Hixson High School who are deaf or hard of hearing. Many classroom and school wide alerts are currently delivered only through sound, which forces deaf and hard of hearing students to rely on teachers or peers. This dependence can affect safety, independence, and overall classroom experience.

The goal of the LED Strobe Light Bell team is to design and build a working prototype that uses bright LED strobe lights to communicate class period changes and other important classroom events through visual cues only. The device is controlled by a programmable microcontroller and is intended to be easy to store, simple to configure, and safe for long term classroom use.

Needs and Criteria

Students who are deaf or hard of hearing need a way to receive classroom and schedule information without relying on others. The device must work as a strictly visual alert system, must not introduce distracting noise or vibrations, and must be safe and easy for staff to operate.

From meetings with the sponsor and teachers, and from prior research on assistive devices, the LED Strobe Light Bell must:

- Provide clear visual alerts that are bright and noticeable from typical classroom locations.
- Operate as a strictly visual system, not using sound or vibration as the primary alert.
- Allow programming of daily and special schedules.
- Be safe to touch and interact with, with no sharp edges or exposed fasteners.
- Be light enough for teachers or staff to move and mount without difficulty.
- Meet relevant safety and accessibility expectations for classroom alert devices.

The device must satisfy several key functions. It must:

- Use visual cues to signal events.
- Read and follow a programmed schedule.
- Switch the strobe lights on and off using a control system and power electronics.
- Enable hanging and sitting position.

- Provide reliable alerts and maintain functionality once device is programmed.

The main objectives of the LED Strobe Light Bell are to be safe to use, easy to understand, simple to program, reliable in operation, and exclusively visual. This report describes how the final design meets those functions and objectives.

Background

Existing devices for deaf and hard of hearing users include assistive listening systems, mass notification platforms, and consumer level products such as wristbands and smart home devices. These systems can improve access to information, but many are either not strictly visual, are expensive to scale to a full school, or are not tailored to classroom use. Prior work on the Hixson High School Visual Alert System identified the need for a low cost, classroom level visual alert that can be installed where it is needed most and that can operate independently of school wide audio systems.

The LED Strobe Light Bell builds on this earlier work. It combines a compact enclosure, high brightness strobe lights, and a programmable microcontroller to create a visual bell that can be adapted to different classrooms and schedules. This report documents the design process, the device functions, the materials used, and recommendations for future improvement.

LED Strobe Light Bell Design and Operation

The LED Strobe Light Bell is housed in a protective enclosure. A microcontroller, for example a Raspberry Pi, reads a stored schedule and controls two 12 V LED strobe lights through a relay module. When the current time matches a scheduled event, the controller activates the strobes for a set duration.

The design includes:

- A control board for schedule storage and timing.
- A relay circuit to switch the higher current strobe lights.
- Two bright LED strobe lights.
- A power supply sized for the controller and strobes and Raspberry Pi.
- A case that is safe for students, keeps wires enclosed, and allows teachers to install and remove the device easily.

The device is intended to require minimal daily interaction. Staff can update the schedule when needed, then allow the system to run automatically. The design section of the report provides more detail on the final configuration.

It is important to note that the internal electronics and software have been fully prototyped and tested, but the final enclosed assembly has not yet been constructed at the time of submitting this report.

I. Introduction

Purpose of Study and Project

The purpose of this project is to create a classroom visual alert device for Hixson High School that supports students and teachers who are deaf or hard of hearing. The device must operate as a strictly visual bell and must present alerts clearly and consistently throughout the school day. It should integrate into typical classroom environments without distracting other students or interfering with instruction.

The device, called the LED Strobe Light Bell, will be prototyped, tested, and documented so that future teams or partners can refine, reproduce, or expand the design.

Purpose of Report

The purpose of this report is to document the complete design process of the LED Strobe Light Bell. This includes:

- Stating the need and describing the problem context.
- Summarizing the collection and evaluation of design options.
- Presenting the final design in enough detail that it can be reproduced.
- Describing the functions, materials, and assembly steps.
- Providing cost information and recommendations for future work.

The report is written so that someone who did not participate in the project can understand the motivation, follow the reasoning behind the design decisions, and rebuild or update the device.

Background

Students who are deaf or hard of hearing often depend on peers and teachers to relay audio only alerts, such as class bells, announcements, or urgent messages. This can affect both safety and independence, which is not sustainable in a classroom environment. Prior work has shown that assistive devices can improve psychosocial outcomes for deaf and hard of hearing individuals when they are accessible and well matched to the user's needs [1]. At the same time, many individuals with hearing differences experience higher rates of internalizing challenges such as anxiety and depression, which can be made worse by constant reliance on others for basic information [2].

At Hixson High School, deaf and hard of hearing students can miss audio bells or announcements when staff are busy or not nearby. While some existing technologies can help, they often fall short for classroom use. Examples include:

- Assistive listening systems that transmit a teacher's voice directly to a receiver [7]. These can be valuable for some students, but they are not suitable for users who are fully deaf and they do not meet the requirement for a strictly visual alert.
- Mass notification systems that drive LED signs, phones, and fire alarms from a central dashboard [5]. These can be powerful, but they may require stable networks, can be expensive to deploy, and may be more complex than needed for a single classroom.
- Consumer devices such as smart doorbells, vibrating alarm clocks, and wearable bands [3, 12]. These are often designed for home use, may rely on phones or personal apps, and can be costly or distracting when scaled to a classroom.

Hixson High School needs a classroom level device that:

- Uses only visual cues to communicate alerts.
- Is simple enough for teachers and staff to operate.
- Is affordable enough for multiple units.
- Can be programmed to follow the school schedule and occasional special events.
- Meets safety and accessibility expectations for a school environment.

Existing Devices and Design Inspiration

Several existing technologies informed the design of the LED Strobe Light Bell.

FM assistive listening systems transmit the teacher's voice to a receiver worn by the student [7, 8]. These systems are well studied and beneficial but are not suitable for a strictly visual solution and do not address visual schedule alerts directly.

Fire alarm systems already provide visual strobe indicators to meet accessibility standards [6, 9]. For example, visible alarm devices must use specific flash rates and colors to remain effective and safe [6]. Although fire alarms are reserved for emergencies and are not appropriate for daily class changes, they illustrate how strobe lights can be used clearly and safely in public buildings.

Commercial visual alert devices such as the Algo 8138 IP Color Visual Alerter provide programmable, networked, multicolor light based notifications [10]. Wearable pagers such as the HomeAware 360 Vibe combine light, vibration, and audio to keep users informed [11]. Consumer starting points such as smart wrist alarms and home signaling devices show how users can benefit from visual and tactile alerts [3, 12]. However, many of these tools either rely on vibration or audio, depend heavily on network connectivity, or cost more than is practical for multiple classrooms.

Programmable hardware platforms such as Raspberry Pi and FPGA based systems offer a flexible way to implement timing logic, control LED outputs, and update schedules [4]. These devices provided useful mental models for creating a visual bell that can be reprogrammed without requiring deep hardware changes.

Taken together, these examples support the need for a classroom level, cost effective, strictly visual device tailored to the schedule and environment at Hixson High School.

Constraints and Assumptions

We had to follow several constraints to make sure our design was safe, reliable, and practical for classroom use.

The device had to weigh less than 10 pounds so that it could be moved and installed easily. We put round edges on the device to prevent injury and make sure there were no components such as screws or nails poking out. The system could only use visual cues, and make sure the device was not too distracting to other students.

During the design process, we made several assumptions. We assumed that the classroom they were in would have a standard wall outlet for power. We also assumed that the teachers would be able to use our device without requiring special training. We assumed that the classroom lighting would change throughout the day and/or during cloudy weather. With these assumptions, we had to make sure the strobe light would still be visible no matter what the lightning is. We expected the device to be able to run for long periods of time without overheating or losing brightness. We also assumed that the Raspberry Pi would be able to store the data for the schedules. These constraints and assumptions helped us create the design for the device that would keep the design simple, safe, and reliable.

II. The Design Decision

Overview of Design Selection

The LED Strobe Light Bell design grew out of earlier concept work for a visual alert system at Hixson High School. That work produced many design options, then narrowed them through constraints, objectives, and comparison tables. For this project, the team built on those ideas and focused on a device that could be realistically fabricated in the freshman engineering laboratory, tested within the course timeline, and used as a prototype for future deployment.

Collection of Possible Design Options

Through individual sketches and team brainstorming, the group considered multiple ways to deliver classroom visual alerts. Some designs were adapted directly from prior concept work, such as box style strobes and displays, while other designs were new variations that used different mounting or control strategies. Online research into other possible devices, brainstorming strategies (such as Heuristics), and other general brainstorming methods were used to create these possible designs.

Constraints, Functions, and Objectives

To compare the designs consistently, the team organized the design requirements into constraints, functions, and objectives.

Constraints

The main constraints are:

- Device must have rounded corners and edges
- Device must not be made from hazardous materials
- Device must not exceed 10lbs
- Device must be free of exposed nails or screws, sharp edges or corners, staples, protruding bolts, blades, splintered wood, and metal burrs
- Device cannot communicate alerts with any method other than visual cues

Functions

The primary functions identified include:

- Provide Alert (Must be strictly visual)
- Accept Programming (Store weekly schedules in order to preserve reliability)
- Accept Power (Power both the Raspberry Pi and the strobe lights)
- Ensure Portability (Ensure device can be easily transported from room to room)
- Ensure Reliability

A function tree that shows the relationships among these functions can be found in Appendix C.

Objectives

The key objectives are:

- Be safe to use.
- Be easy to transport
- Be exclusively visual
- Be user friendly (For teachers and other staff)
- Be reliable (Operate consistently throughout the school year)
- Be inexpensive to manufacture

An objectives tree that breaks these goals into measurable sub objectives is also included in Appendix C.

Decision Process

1. Introduction
   This section communicates the final design selection for the LED Strobe Light Bell and

the reasoning behind that decision. It summarizes the initial collection of possible design concepts, the decision making tools and evaluation process that were applied, the constraints and functional requirements that influenced the choices, and why the selected design best meets the project objectives and constraints.

2. Collection of Possible Design Options
   Through individual and team brainstorming sessions, the team created 33 possible design sketches for this project. These designs were developed using various techniques, including online research, solo ideation, team ideation, and analyzing functions, objectives, and constraints, as well as other general brainstorming methods. One of the most important constraints for this project was that the device must be strictly visual and must use no other method for alerting the students and teachers.

3. Constraints, Functions, and Objectives
   After looking at all the possible designs, the team removed options that did not fulfill the objectives or did not meet the constraint and function requirements.

   - Constraints: cost effective, safe to use, user friendly, strictly visual, consistent with school safety expectations, and does not disrupt other students.
   - Functions: use visual cues, ensure programmability, communicate classroom events, allow the device to be transported and mounted, remember commands or schedules, and display or signal information.
   - Objectives: be easy to transport, be easy to understand, be exclusively visual, be simple to use, be reliable, and be simple to program.

4. Decision Process

   1. Initial Brainstorm
      The team started with 33 possible solutions before any comparison or reduction. These sketches came from each individual team member's ideas and from sketches completed as a team using several brainstorming techniques.

   2. Combination, Functions, and Constraints
      As part of this screening process, several designs were removed from consideration because of redundancy, combination with similar ideas, or failure to meet the functional or constraint requirements:

      - Some concepts were combined with team sketches due to overlapping features, and others were eliminated because they did not satisfy cost effectiveness and portability requirements.
      - Designs that lacked portability, included sharp edges that posed safety concerns, or would have exceeded budget limits were removed.
      - Concepts that required excessive wall space or permanent building changes were also removed.

After this refinement process, 16 viable design options remained. These included all team members' group sketches, plus several individual designs that showed strong alignment with the required functions and constraints.

5. Device Comparisons
   After determining the 16 remaining possible solutions, the team compared them using four comparison tables, shown in Appendix C. Each device was given a rating of poor, adequate, or exceptional for each of the objectives. This process left one superior design for each of the comparison tables.

6. Pairwise Comparison Table and Weighted Objective Tree
   After considering the functions of each device with the comparison tables, the team selected a total of four remaining devices. To further eliminate designs and possible solutions, a more detailed method was used. A pairwise comparison table was created and then used to create a weighted objective tree. This process allowed the team to weight each objective based on how important it was with respect to the others, as shown in the figures in Appendix C.

7. Evaluation Table
   After creating the weighted objective tree, the team constructed a weighted evaluation table to narrow the selection to just one final solution. The table scores each device based on every objective, with the importance of that objective taken into consideration. The resulting total scores showed which designs best balanced performance with the weighted objectives.

8. Basis for Final Selection
   After evaluating the final four design concepts and comparing their performance against the established functions, objectives, and constraints, two designs clearly stood out as the most effective: the Suction Cup LED Display Board and the Strobe Light LED Box.

9. Final Selected Design
   Overall, the Strobe Light LED Box was selected due to cost efficiency and issues that could be present with using a suction cup system to store and mount the device within the classroom. After the final device was selected, a better name was chosen and was named the LED Strobe Light Bell.

The LED Strobe Light Bell is a compact visual alert device sized for classroom use. It features an enclosure with rounded edges for safety and easy handling. The design incorporates high brightness strobe lights for attention grabbing visual alerts, controlled by a Raspberry Pi that also manages the alert schedule. The device provides programmable alerts through the use of strobe lights to enhance visibility and help ensure that the signal can be seen from typical locations in the room.

10. Conclusion
    Through a structured and detailed design selection process, the team successfully narrowed an initial list of 33 ideas down to one final, optimized solution: the LED Strobe

Light Bell. Each stage of the process, from the elimination of redundant or non functional concepts to the weighted evaluation and pairwise comparison analysis, was guided by the project's key constraints, functions, and objectives.

III. The Design

LED Strobe Light Bell Customer and User Objectives

The most important objective of the LED Strobe Light Bell is that it can be used safely in a high school classroom. The device must be free of exposed screws and wiring, must not have any sharp edges or 45 degree angles, and should not introduce any new hazards during normal classroom activity.

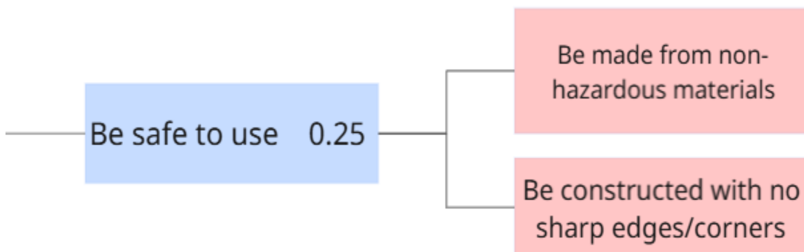


Figure 1 - Objective Tree Branch (Be Safe to Use)

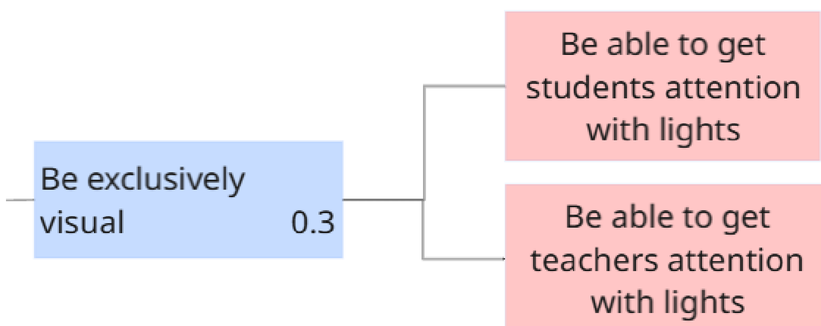


Figure 2 - Objective Tree Branch (Be Exclusively Visual)

The above figures show the most important objectives within the objective tree. The importance of these objectives was determined using a pairwise comparison table as mentioned in the section above.

To meet the safety objectives, all external corners and edges of the enclosure will be rounded or smoothed, and any mounting hardware will either be recessed or covered. Internal wiring will be fully enclosed within the housing so that students and staff cannot contact live conductors. If any fasteners extend through the enclosure, they will be trimmed and covered so that they cannot scratch or snag.

Figure 3 shows the exterior of the LED Strobe Light Bell enclosure highlighting rounded edges and enclosed hardware.

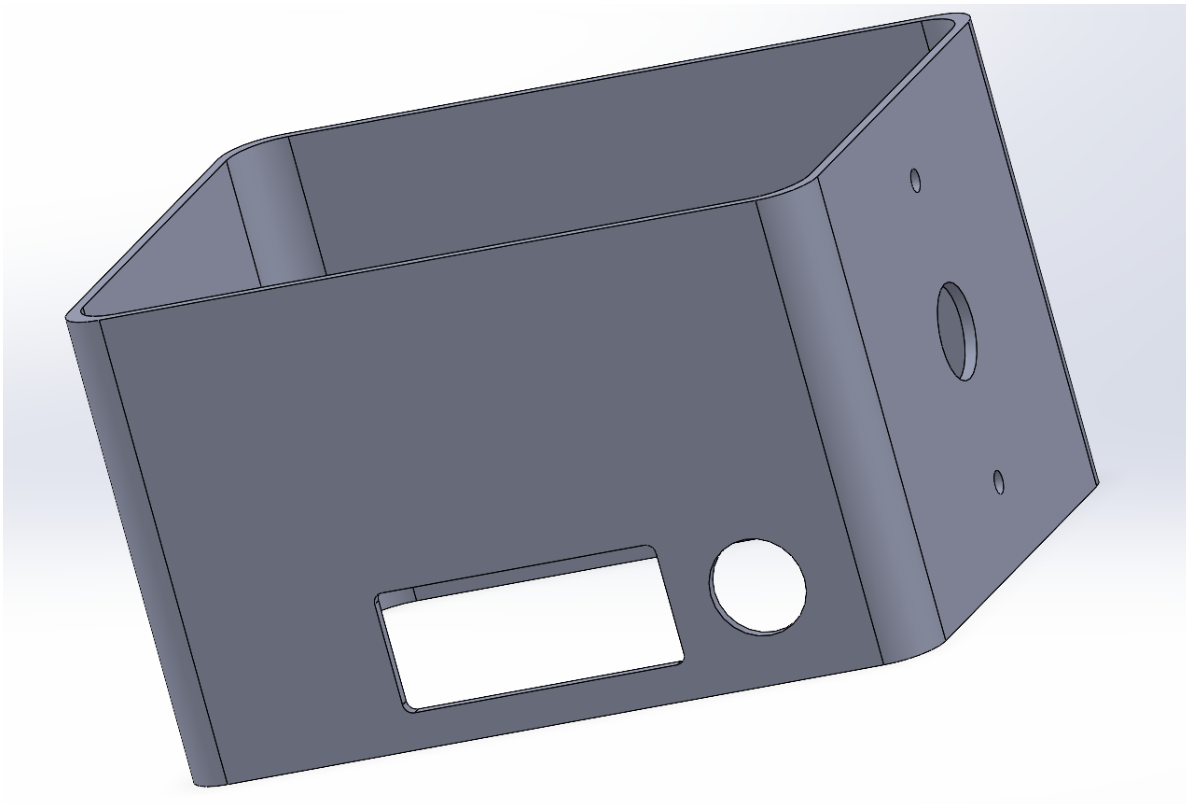


Figure 3 - CAD Drawing of Enclosure

The figure above shows a CAD drawing of the enclosure. The enclosure contains mounting holes to attach the LED Strobe Lights to the side of the case, as well as precisely cut holes for the Raspberry Pi and power cables to be routed through for simple connection and use.

In addition to safety, the device must provide clear and reliable visual alerts. The strobe output must be bright enough and positioned so that students can reliably see the signal from typical seating positions across the room, without being uncomfortably intense when viewed at close range. The device uses only visual cues, with no audio or vibration, to meet the requirement of a strictly visual alert for students who cannot access sound.

The figure below shows an image of the device fully wired and connected. (At the time of making this report the enclosure has not yet been made)

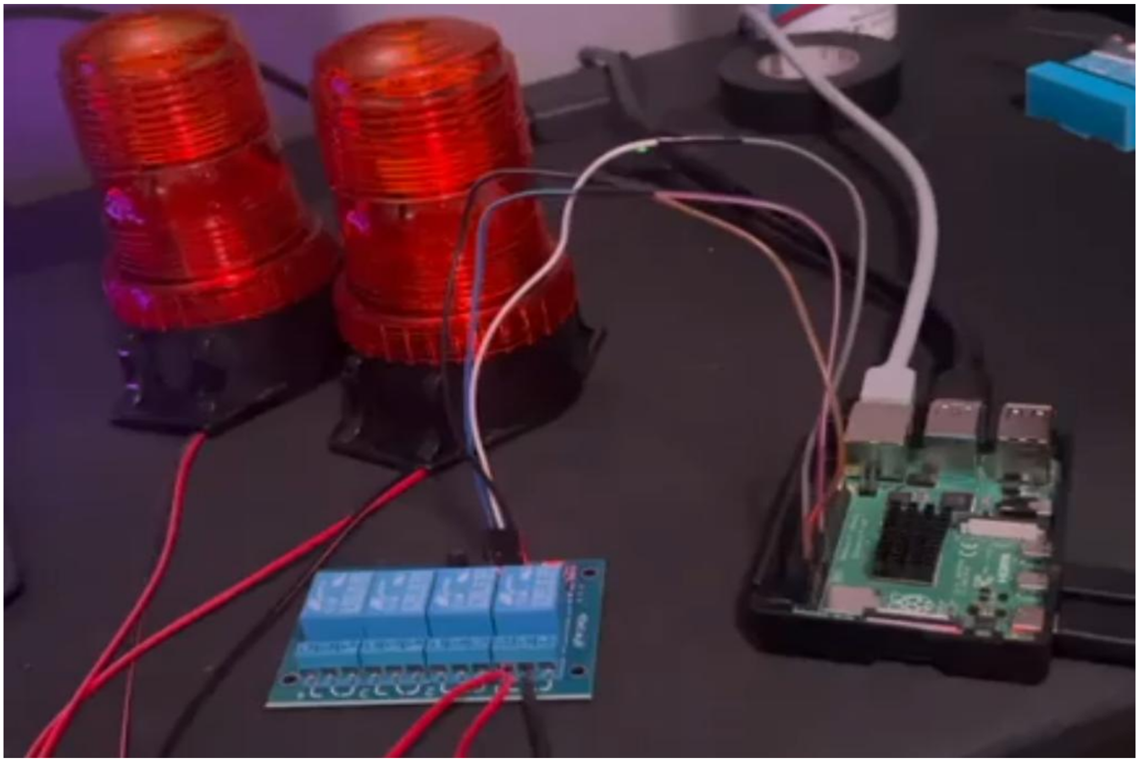


Figure 4 - Fully Functional and Wired Device (No enclosure)

Ease of use is another key objective. Teachers and staff should be able to understand the status of the device at a glance, turn it on and off when needed, and update bell schedules without needing advanced programming knowledge. The figure below shows the simple program that was created to ensure that the device could be used by anyone regardless of background knowledge.

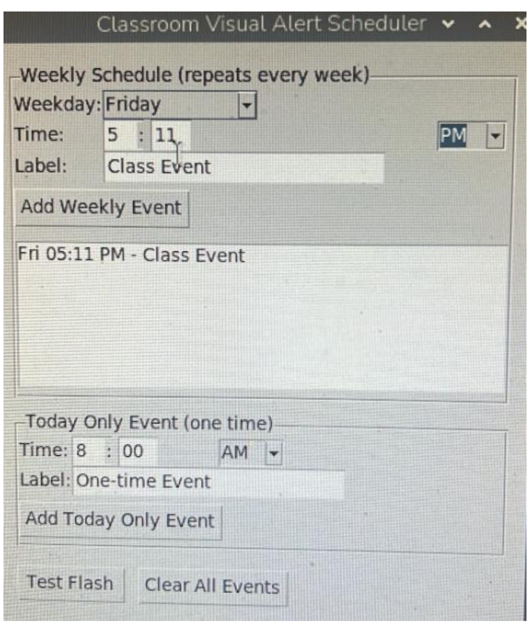


Figure 5 - GUI Created for Device Control

This simple GUI features a Weekly Schedule and Today Only Event scheduler. It works by storing the weekly events that are created in a file that saves to the Raspberry Pi. This makes sure that the device remains functional after reboots, device crashes, and power outages. This functionality was tested overnight through several device reboots and proved to be successful. An additional prototyping test was conducted where a user with no previous knowledge was tasked with creating a daily and weekly alert. The test was successful, and the alerts were created within a few minutes of using the application.

LED Strobe Light Bell Device Functions

Before fabricating the final device, the team identified required, supplemental, and unwanted functions. A function node tree was used to narrow down these functions and is included in the figure below.

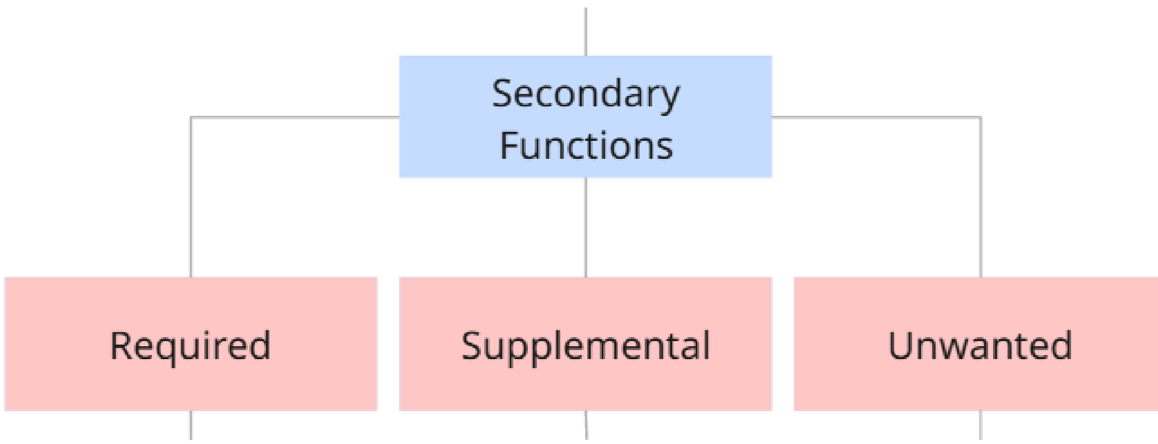


Figure 6 - Function Node Tree (Secondary Functions)

Required Functions

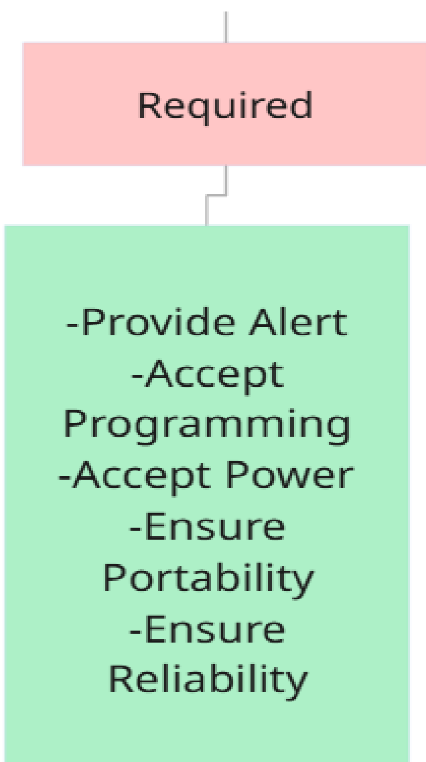


Figure 7 - Required Functions

The required functions describe the functions that are essential for the device's operation. This means that if the final prototype did not satisfy these requirements, then the device was incomplete.

The required functions for the LED Strobe Light Bell are:

- Provide Alert
- Accept Programming
- Accept Power
- Ensure Portability
- Ensure Reliability

To satisfy these functions, the team selected:

- A pair of 12 V amber colored LED strobe lights
- A Raspberry Pi that was programmed with a simple GUI that provides ease of use for staff and teachers at Hixson High School
- A 12 V DC wall adapter, USB-C power adapter for the Raspberry Pi, and an extension cord that was connected to the wall
- A 3D printed PLA plastic enclosure with routing holes and mounting spots for the LED Strobe lights to be attached to each side
- Each of the above parts and devices proved to be reliable among a series of tests

Supplemental Functions

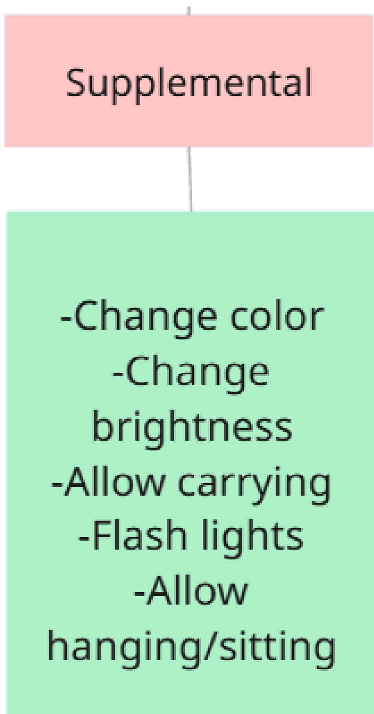


Figure 8 - Supplemental Functions

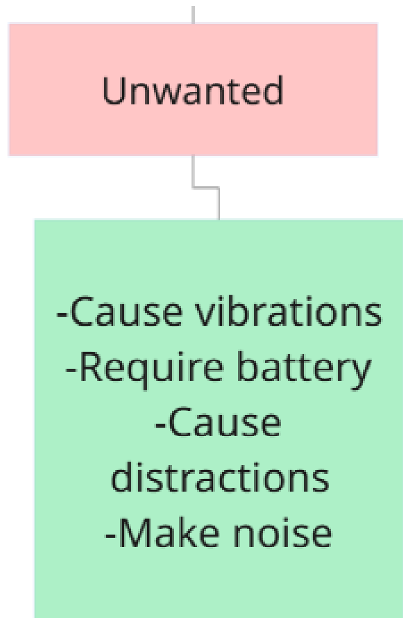
Supplemental functions were determined based on consolation with the client. These are not strictly required, however can improve device functionality.

Several supplemental functions were included to improve usability even though they are not strictly required for basic alert operation. These include:

- Change Color
- Chang Brightness
- Allow carrying
- Flash Lights
- Allow hanging/sitting position

Three of the supplemental functions were able to be met by the final device due to limitations with budget, time, and consultation with the client to ensure they were satisfied with the solution without them. These include allow carrying, flash lights, and allow hanging/sitting.

Unwanted Functions



Unwanted Functions are functions that are preferred to not be in the final solution. Some of these functions could come from specific devices and parts that are selected to satisfy other required or supplemental functions.

The team identified several unwanted functions to avoid negative outcomes. These include:

- Cause vibrations
- Require battery
- Cause distractions
- Make noise

Figure 9 - Unwanted Functions

The final device satisfies all the unwanted functions creating a non distracting, and non battery operated final prototype for ease of use and effortless functionality.

Material Specifications

There are several main components required to create the LED Strobe Light Bell.

The device requires the following parts:

1. Raspberry Pi 4
   Raspberry Pi 4 Model B used as the controller for storing the schedule, reading time, and switching the relay.

2. Two QWORK LED strobe lights
   12 V DC QWORK amber LED strobe lights, each supplied with mounting screws and nuts.

3. Custom 3D printed PLA enclosure
   PLA enclosure designed in SolidWorks to hold the lights, Raspberry Pi, relay module, and internal wiring.

4. 12 V AC to DC power adapter
   Wall adapter that outputs regulated 12 V DC with enough current to power both strobe lights.

5. Raspberry Pi USB-C power adapter
   5 V USB-C power adapter rated for the Raspberry Pi 4.

6. Extension cord
   Grounded extension cord used to supply power to both the Raspberry Pi adapter and the 12 V adapter.

7. Female to female jumper wires
   Three insulated jumper wires for connecting Raspberry Pi GPIO pins to the relay module.

8. Relay module
   Single channel 5 V relay module with contacts rated for switching the 12 V strobe lights.

9. Lamp wire
   Two conductor lamp cord for distributing 12 V power from the adapter to the relay and strobe lights.

Modified Materials Used by Team

Several components were modified to better fit the specific requirements of the project. Examples of these modifications include:

- Adding holes or slots for ventilation while keeping users protected from contact with internal parts.
- Trimming and routing wires to appropriate lengths, then securing them to prevent snagging and to keep the interior organized.

Any machining or cutting operations were carried out using tools in the Freshman Engineering Workshop following appropriate safety procedures. Steps that affect structural integrity or safety, such as drilling new mounting points or cutting openings in the enclosure, were planned carefully to avoid weakening the housing.

Assembly of the LED Strobe Light Bell

The assembly of the LED Strobe Light Bell can be divided into a series of steps, including component preparation and wiring, as seen below:

1. Place the 3D-printed PLA enclosure horizontally on the work surface with the strobe mounting faces positioned on the left and right sides.
2. Insert the 2 QWORK 12 V strobe lights into the pre-modeled mounting holes on each side of the enclosure.
3. Secure each strobe light using the included screws and nuts until firmly tightened.
4. Route the red and black wires from each strobe through the interior wire openings of the enclosure.
5. Tape the wiring entry points with electrical tape to prevent abrasion against the PLA edges.
6. Install 8 PLA standoffs inside the enclosure in the positions designated for supporting the Raspberry Pi and relay module.
7. Attach the Raspberry Pi 4 to four of the standoffs using small machine screws.
8. Attach the single channel 5 V relay module to the remaining four standoffs.
9. Feed the 12 V AC-to-DC adapter's output wire into the enclosure through the power entry slot.
10. Strip approximately ½ inch of insulation from the ends of the lamp wire used to distribute 12 V power.
11. Connect the positive lamp wire to the relay's COM terminal and tighten the screw connector.
12. Connect the relay's NO terminal to both strobe light positive wires by twisting the conductors and securing them with a wire connector or electrical tape.
13. Connect both strobe light ground wires to the 12 V adapter's negative terminal.
14. Connect three female to female jumper wires from the relay input pins to the Raspberry Pi GPIO header (signal, 5 V, and ground).
15. Tape all exposed wire joints with electrical tape to cover bare conductors and ensure safe operation.
16. Organize the wires inside the enclosure to separate low voltage GPIO wiring from the 12 V power wiring.
17. Inspect all electrical connections for correct polarity and tightness.
18. Plug the Raspberry Pi's USB-C power cable into its power port and route the cable out of the enclosure.
19. Place the Raspberry Pi power adapter and the 12 V adapter onto a shared extension cord.
20. Verify once more that all internal components are seated securely and that no wires are pinched between enclosure surfaces.
21. Close the enclosure by placing the top cover onto the base and securing it with screws.
22. Power the Raspberry Pi and 12 V adapter to confirm that the relay activates the strobes when triggered by software.

Operation of the LED Strobe Light Bell

The LED Strobe Light Bell is intended to run mostly unattended once the schedule has been programmed, but there are a few basic operating steps for daily use and configuration.

NOTE: The program will be provided by contacting one of team members and the contact information will be listed at the end of the report.

For daily classroom use:

1. Confirm that the power supply is connected and that the power indicator shows that the device is on.
2. At each scheduled time, the microcontroller activates the relay, which turns on the strobe lights for the preset alert duration.
3. When the alert period ends, the controller switches the strobe lights off automatically.

For schedule updates:

1. Plug in HDMI, keyboard, and mouse into the Raspberry Pi.
2. Plug the other end of the HDMI into a monitor (Will be provided for Hixson HS)
3. Edit the program using the provided intuitive buttons (add weekly schedule, add daily event, and clear weekly schedule)
4. Unplug all cables and set device in preferred classroom location

As well as the LED Strobe Light Bell, an additional smart LED display board will be included and delivered to the Hixson High School Staff. This device ended up being removed from the final device due to integration with the Raspberry Pi. However, when the team met with the client, it was determined that they could still use the display board for more simple classroom notifications.

This device works with an application downloaded to mobile phones and can display a wide variety of messages, different colors and font sizes, as well as store a few presets for daily or out of the ordinary events. An image of this device is included in the appendix.

IV. Cost of the Device Build

We tracked the total cost of all materials and components used to build the Hixson High School Visual Alert System. Our design required several purchased parts and a 3D printed component (the case). The breakdown of the costs is shown below.

Item	Description	Quantity	Cost (USD)
Strobe Lights	LED Emergency strobes used for visual alerts.	2	$14.97
Raspberry Pi	Main controller used to run the schedule and lighting program.	1	$109.99
Relay Module	Used to control the power to the strobe lights from the Raspberry Pi	1	$6.99
Female-to-Female Jumper Wires	Used to connect components to the Raspberry Pi	1 pack	$2.99
3D Printed Outer case	Custom-printed shell created via SolidWorks. Houses all device components.	1	$33.48
LED Display Board	Purchased but not used in the final build.	1	$100.00
Total Project Cost			$268.42

Table 1: Total cost of project before shipping and taxes.

The total cost for all materials that were used in the working prototype is $268.42. If we include the unused LED Display Board, the cost becomes $368.42. This cost stays within the project's budget limit of $400.

V. Closing Material

Conclusions and Recommendations

The LED Strobe Light Bell project demonstrates how a structured engineering design process can be used to create a practical, safe, and effective visual alert system for deaf and hard-of-hearing students at Hixson High School. Beginning with the need for a strictly visual classroom alert, the team evaluated more than thirty design concepts, narrowed them through constraints and objective comparisons, and produced a prototype that meets the essential requirements of clarity, safety, and reliability. The final device uses two LED strobe lights controlled by a Raspberry Pi running a simple scheduling interface. Testing showed that the strobes activate consistently, remain visible under different lighting conditions, and operate reliably across repeated daily cycles. The software was also tested through several reboots and outages, confirming that the schedule is preserved and restored automatically. Safety goals were met by enclosing all wiring, preventing exposed conductors, and ensuring that no sharp edges or hazardous features were present in the design.

Overall, the LED Strobe Light Bell meets its intended purpose, providing a dependable and accessible visual alert for students who cannot rely on sound or other students. It also proved simple for teachers and staff to operate, and its modular structure allows future groups to continue developing or expanding the system. The design offers a strong foundation for a classroom ready device that can improve safety, independence, and equal access to information for students at Hixson High School.

NOTE: *While all electronic subsystems and software components have been tested and verified, the final enclosed assembly has not yet been completed. The conclusions reflect the performance of the functional prototype without its enclosure.*

Recommendations

Based on testing, user needs, and the prototype's performance, the team recommends the following improvements for future versions of the device:

1. **Add a Safe and Flexible Mounting System**
   Future iterations should include a lightweight, classroom-safe mounting method to position the device consistently and improve visibility during alerts.

2. **Integrate a Programmable LED Display Board**
   Adding a small programmable display controlled by the Raspberry Pi would allow the device to show short visual messages alongside the strobe alerts, improving communication while remaining fully visual.

3. **Expand the Existing GUI Application**
   The current GUI could be made more accessible by adding schedule previews, clearer prompts, and user-friendly features such as error messages or adjustable settings.

VI. References

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[10] Algo Communication Products Ltd., “8138 IP Color Visual Alerter,” Online.

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Update the formatting to match what your instructor prefers and add any additional sources that specifically describe your hardware or programming resources.

[13] OpenAI, 2025, *ChatGPT* — Large Language Model used for writing assistance, structural organization, and formatting guidance for many portions of this report.

VII. Acknowledgements

The group would like to thank the following people for their support and contributions to the LED Strobe Light Bell project.

- **Dr. Cecelia Wigal**, for sponsoring the project, offering consistent guidance throughout the semester, and helping the team interpret project requirements in a realistic and achievable way. Her feedback during check-ins and design reviews played an essential role in shaping the final direction of the device and ensuring that the team stayed focused on meaningful engineering objectives.

- **Hixson High School staff**, for describing the daily classroom needs of deaf and hard-of-hearing students and providing critical context about how visual alerts are used in practice. Their input allowed the team to better understand the functional challenges students face and to design a device that supports independence and accessibility. The team is also very grateful for the staff being extremely understanding of the issues we encountered during the construction process, as well as for offering many excellent recommendations to ensure the device meets their needs.

The group extends sincere appreciation for the time, patience, and collaboration provided by everyone involved. Their expertise and willingness to share real-world experiences greatly informed the design decisions made throughout the project. The LED Strobe Light Bell could not have been completed without their continued encouragement, insight, and support.

Appendix A: Drawings

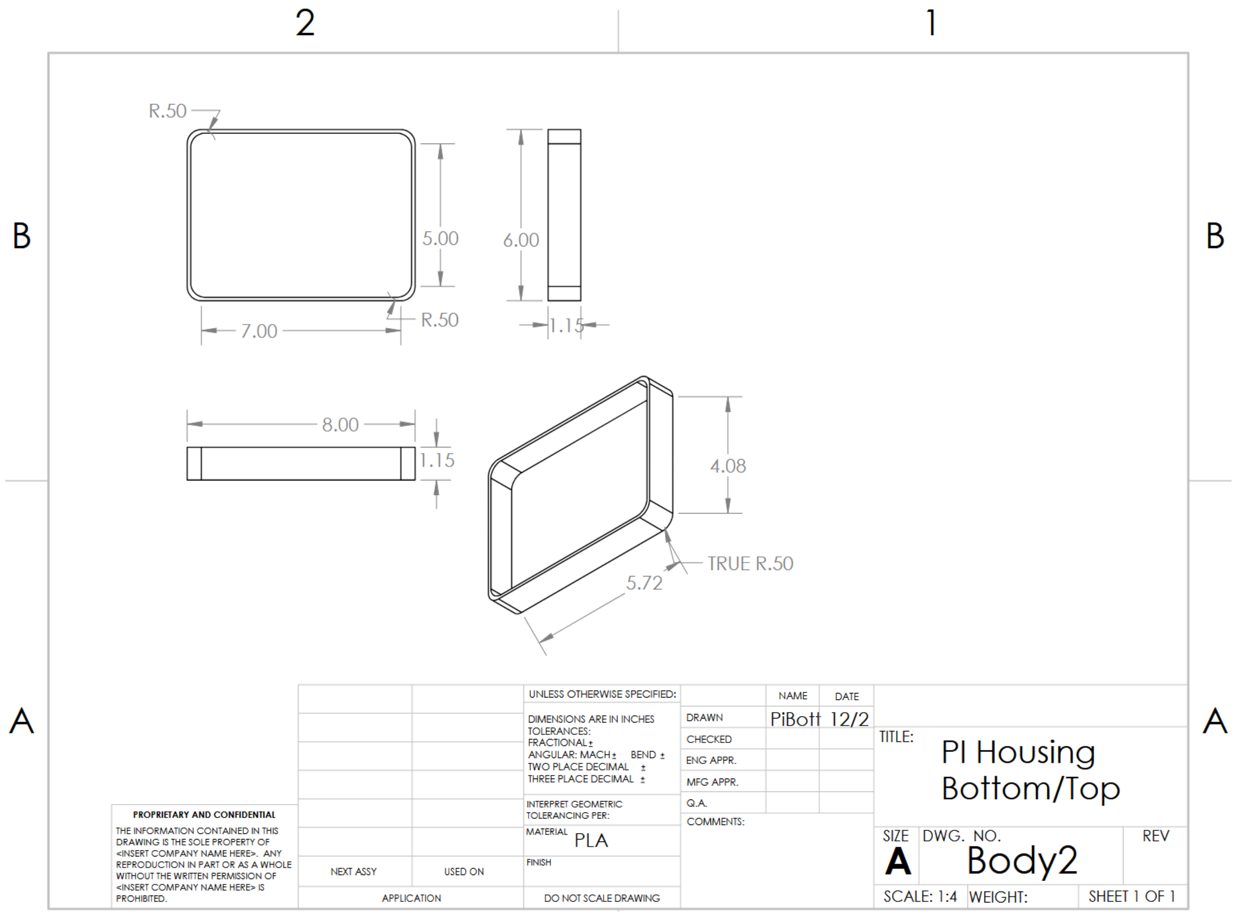

Figure 10 – Enclosure Dimensions (Bottom / Top)

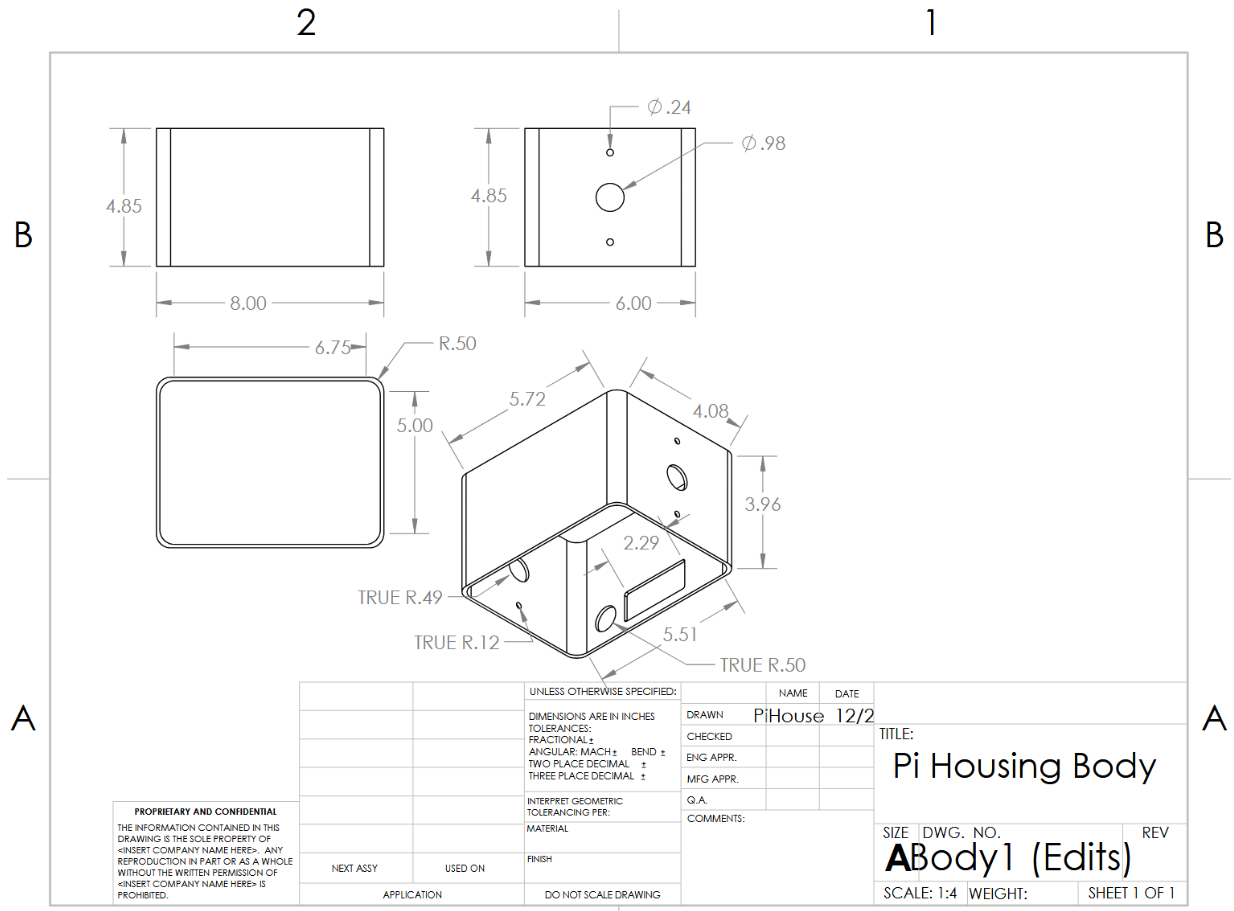

Figure 111 – Enclosure Dimensions (Body)

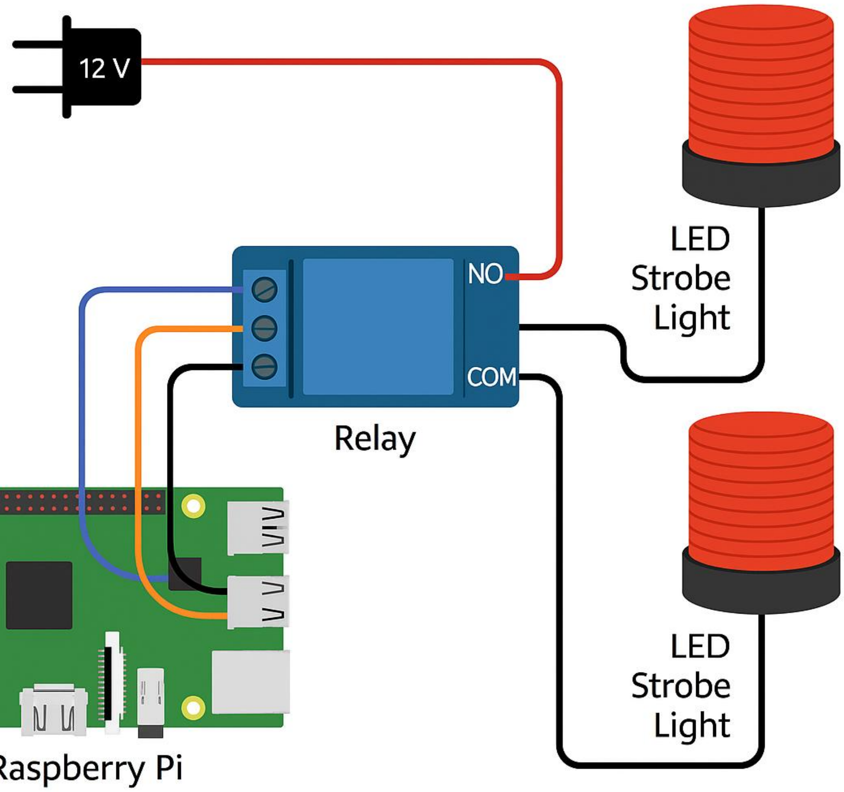

Figure 12– Wiring Diagram

Appendix B: Build Instructions

This appendix provides step-by-step instructions for constructing the LED Strobe Light Bell using handheld tools and the required components. These instructions are written so that the device can be recreated by future teams using the same methods and tools available in the Freshman Engineering Studio.

The following handheld tools are required:

1. One pair of wire strippers
2. One flat head screwdriver
3. One Phillips head screwdriver
4. One bottle of general purpose glue suitable for PLA plastic
5. One roll of electrical tape

Because the enclosure is fully modeled and printed in CAD, **no drilling or cutting tools are required** for assembly.

The main components needed to construct the LED Strobe Light Bell are listed below. These items can be purchased from online retailers such as Amazon or electronic supply vendors:

1. One Raspberry Pi 4 Model B
2. Two QWORK 12 V amber LED strobe lights (mounting screws and nuts included)
3. One custom 3D-printed PLA enclosure with pre-modeled openings for wiring, ports, and mounting points
4. One 12 V DC wall adapter for powering the strobe lights
5. One 5 V USB-C Raspberry Pi power adapter
6. One grounded extension cord
7. Three female-to-female jumper wires
8. One generic single-channel 5 V relay module
9. One length of two-conductor 18 AWG lamp wire

With the required tools and parts gathered, construction of the LED Strobe Light Bell can begin. The enclosure and internal electronics are assembled in the order shown below.

Assembly of the LED Strobe Light Bell

1. Place the fully printed PLA enclosure on the work surface with the strobe mounting faces positioned on the left and right sides.
2. Insert each of the two QWORK 12 V strobe lights into their pre-designed mounting holes on the enclosure.
3. Secure each strobe light using the included screws and nuts until they are held firmly in place.
4. Route the red (positive) and black (ground) wires from each strobe light into the enclosure through the pre-modeled wire channels.
5. Apply electrical tape around the wire entry edges to prevent abrasion against the PLA surface.
6. Identify the eight interior PLA standoffs: four designated for the Raspberry Pi and four for the relay module.
7. Position the Raspberry Pi 4 onto its four standoffs and secure it using the appropriate screws.
8. Position the single-channel 5 V relay module onto its four standoffs and secure it with screws.
9. Feed the output cable of the 12 V DC power adapter through the enclosure’s rear power slot.

10. Strip approximately ½ inch of insulation from the ends of the lamp wire using wire strippers.
11. Connect the positive conductor of the lamp wire to the relay's COM terminal and tighten the screw terminal.
12. Join the relay's NO terminal to both strobe light positive wires by twisting the wires together and securing the connection tightly with electrical tape.
13. Connect both strobe light ground wires directly to the negative terminal of the 12 V adapter output, covering the joint fully with electrical tape.
14. Attach three female-to-female jumper wires from the relay input pins to the Raspberry Pi GPIO header, connecting signal, power, and ground as required by the device software.
15. Wrap electrical tape around all exposed wire junctions to prevent accidental contact and to ensure safe operation.
16. Organize all internal wiring so that low-voltage GPIO wiring remains separated from the higher-current 12 V wiring.
17. Inspect all connections to verify correct polarity, secure attachment, and proper routing.
18. Connect the Raspberry Pi's USB-C power cable and route it through the pre-modeled access opening in the enclosure.
19. Place both the Raspberry Pi 5 V adapter and the 12 V adapter onto a shared extension cord to prepare for power testing.
20. Ensure that no internal wires are pinched or resting against the screw posts where the enclosure will close.
21. Glue the lower enclosure section together following the printed seam lines and allow it to dry fully.
22. Place the top enclosure cover onto the glued base and secure it with screws so that the device can be easily reopened for maintenance.
23. Perform a functional test by powering the Raspberry Pi and the 12 V adapter to confirm that the relay switches correctly and that the strobe lights activate when triggered by the scheduling software.
24. After confirming correct operation, the device is ready for placement in a classroom setting.

Appendix C: Criteria Figures

Objective tree, function tree, function node tree, pairwise comparison tables, and weighted evaluation tables.

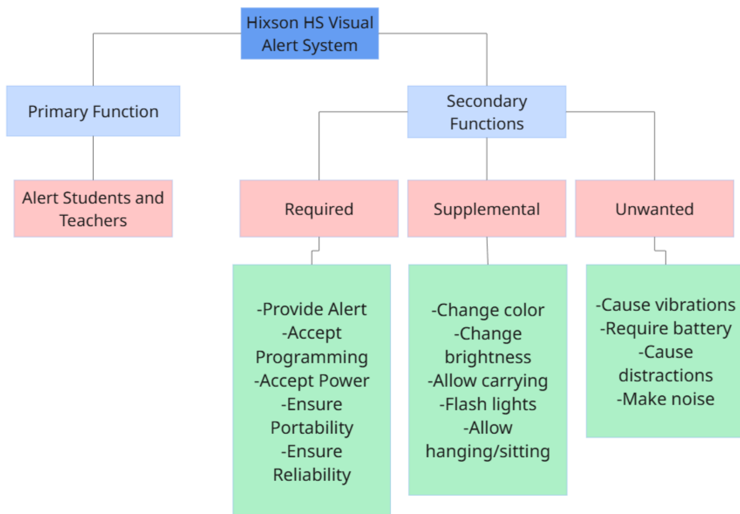

Figure 13 – Function Node Tree

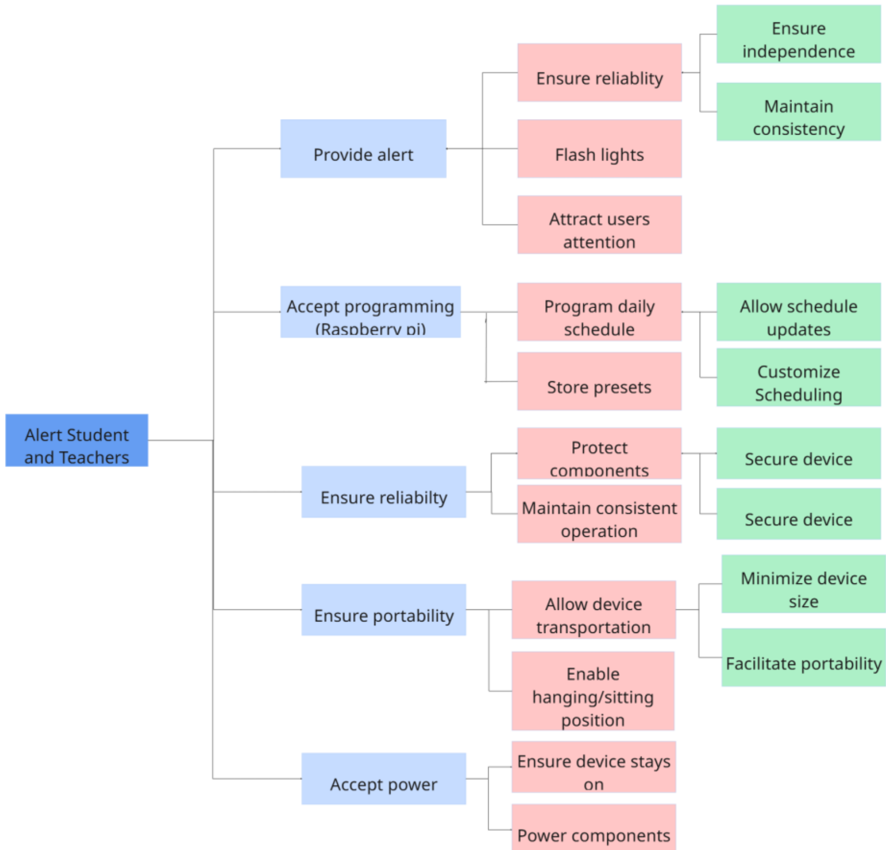

Figure 14 – Function Tree

	Be Strictly visual	Be easy to transport	Be user friendly	Be inexpensive	be reliable	be safe to use	be customizable	Total	Normalized Importance
Be strictly visual	N/A	1	1	1	1	1	1	6	0.3
Be easy to transport	0	N/A	0	1	0	0	1	2	0.1
Be user friendly	0	0	N/A	0	1	0	1	2	0.1
be inexpensive	0	0	1	N/A	0	0	1	2	0.1
be reliable	0	1	1	0	N/A	0	1	3	0.15
be safe to use	0	1	1	1	1	N/A	1	5	0.25
be customizable	0	0	0	0	0	0	N/A	0	0
								20	1

Figure 15 – Pairwise Comparison Table

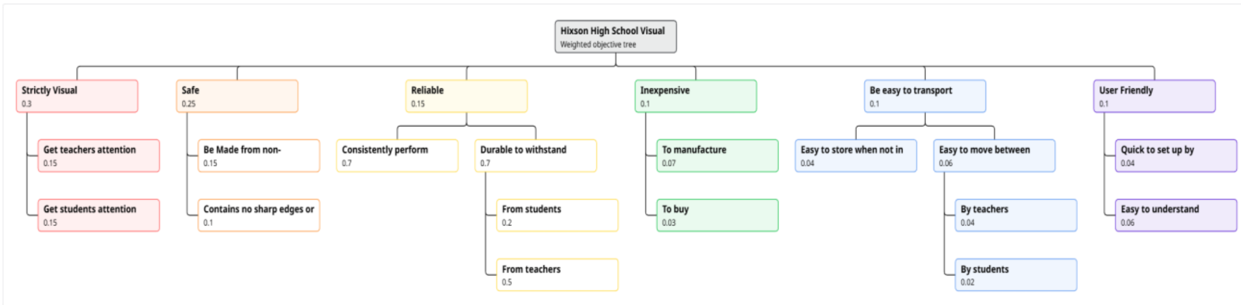

Figure 16 – Weighted Objective Tree

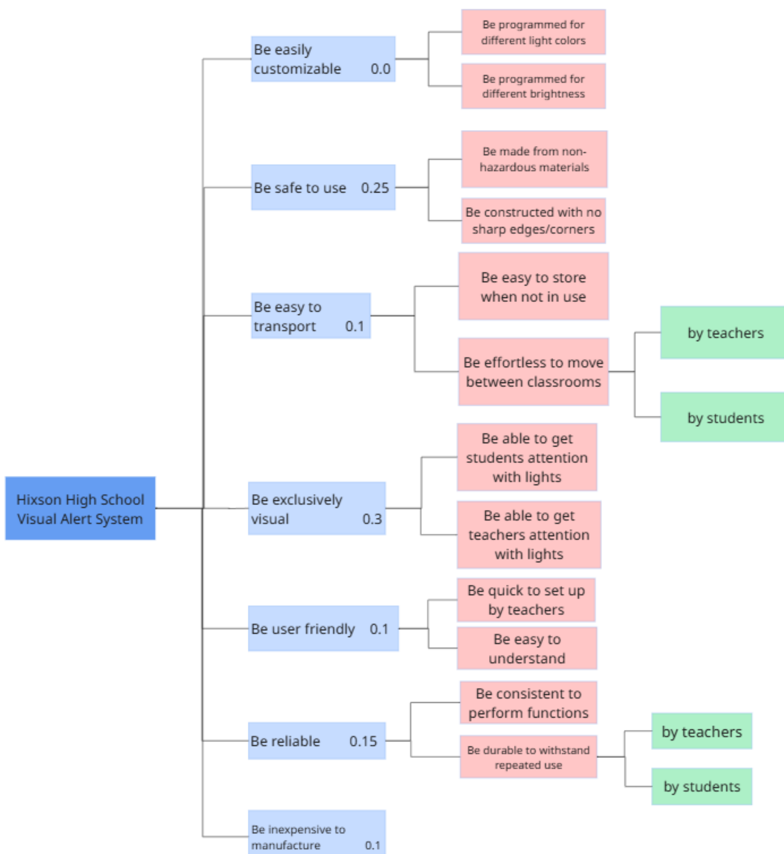

Figure 17 – Objective Tree

Design Options #1

	Suction Cup LED Screen	Projector System	Rolling Cart Display	Rolling Smart Board TV
Be safe to use	Excellent	Excellent	Adequate	Adequate
Be reliable	Adequate	Poor	Adequate	Excellent
Be Strictly Visual	Excellent	Adequate	Excellent	Excellent
Be Easy to Transport	Excellent	Poor	Adequate	Excellent
Be Customizable	Excellent	Poor	Poor	Poor
Be user friendly	Excellent	Adequate	Adequate	Adequate
Be Inexpensive	Excellent	Adequate	Adequate	Poor
TOTAL	20	17	14	15

Design Options #2

	Door Mounted LED Panel	Portable Alarm Box	Rolling Projector	Window LED Light Strip
Be safe to use	Excellent	Excellent	Excellent	Adequate
Be reliable	Adequate	Adequate	Aqequate	Adequate
Be Strictly Visual	Excellent	Excellent	Excellent	Excellent
Be Easy to Transport	Poor	Excellent	Excellent	Excellent
Be Customizable	Poor	Poor	Adequate	Adequate
Be user friendly	Adequate	Adequate	Excellent	Adequate
Be Inexpensive	Adequate	Excellent	Excellent	Adequate
TOTAL	14	17	19	16

Design Options #3

	Classroom Light Interceptor	Strobe Light LED Box	Clock Display System	Door Mounted LED Strip (Heuristics)
Be safe to use	Poor	Excellent	Excellent	Excellent
Be reliable	Poor	Adequate	Poor	Excellent
Be Strictly Visual	Excellent	Excellent	Adequate	Excellent
Be Easy to Transport	Poor	Excellent	Poor	Poor
Be Customizable	Excellent	Adequate	Poor	Excellent
Be user friendly	Poor	Adequate	Poor	Excellent
Be Inexpensive	Poor	Excellent	Excellent	Excellent
TOTAL	11	18	12	19

Design Options #4

	Rolling Smart TV Display (Heuristics)	Audio Display Box	Beacon Flash LED Unit	Portable Projector Box
Be safe to use	Adequate	Excellent	Excellent	Excellent
Be reliable	Excellent	Adequate	Excellent	Poor
Be Strictly Visual	Excellent	Adequate	Excellent	Excellent
Be Easy to Transport	Excellent	Excellent	Adequate	Excellent
Be Customizable	Adequate	Excellent	Excellent	Poor
Be user friendly	Excellent	Adequate	Adequate	Adequate
Be Inexpensive	Poor	Poor	Poor	Excellent
TOTAL	15	15	17	16

Figure 18 - Comparison Tables from Final 16 Devices

		Suction cup LED Board			Door Mounted Led Strip			Rolling Projector			Strobe light LED Box		
Objective	Relative Weights	Score	Value	Weighted Value	Score	Value	Weighted Value	Score	Value	Weighted Value	Score	Value	Weighted Value
Get teachers attention using lights	0.15	Excellent	3	0.45	Excellent	3	0.45	Excellent	3	0.45	Excellent	3	0.45
Get students attention using lights	0.15	Excellent	3	0.45	Excellent	3	0.45	Excellent	3	0.45	Excellent	3	0.45
Made from non-hazardous material	0.15	Excellent	3	0.45	Excellent	3	0.45	Excellent	3	0.45	Excellent	3	0.45
No sharp edges or corners	0.1	Excellent	3	0.3	Excellent	3	0.3	Excellent	3	0.3	Excellent	3	0.3
Consistently perform functions	0.07	Excellent	3	0.21	Adequate	2	0.14	Adequate	2	0.14	Excellent	3	0.21
Durable to withstand epeated use rom teachers	0.05	Excellent	3	0.15	Adequate	2	0.1	Adequate	2	0.1	Excellent	3	0.15
Durable to withstand epeated use rom students	0.02	Excellent	3	0.06	Adequate	2	0.04	Adequate	2	0.04	Excellent	3	0.06
Inexpensive to nanufacture	0.07	Excellent	3	0.21	Excellent	3	0.21	Poor	1	0.07	Excellent	3	0.21
Inexpensive to buy	0.03	Excellent	3	0.09	Excellent	3	0.09	Poor	1	0.03	Excellent	3	0.09
Easily stored when not in use	0.04	Excellent	3	0.12	Poor	1	0.04	Adequate	2	0.08	Excellent	3	0.12
Easy to move between lassrooms by teacher	0.04	Adequate	2	0.08	Poor	1	0.04	Adequate	2	0.08	Excellent	3	0.12
Easy to move between lassrooms by students	0.02	Adequate	2	0.04	Poor	1	0.02	Adequate	2	0.04	Excellent	3	0.06
asy to set up by teacher	0.04	Excellent	3	0.12	Excellent	3	0.12	Adequate	2	0.08	Excellent	3	0.12
Easy to understand	0.06	Adequate	2	0.12	Excellent	3	0.18	Adequate	2	0.12	Excellent	3	0.18
Total				2.85			2.45			2.43			3.51

Figure 19 - Evaluation Table from Final 4 Devices

Appendix D: Team Member Participation Documentation

Documentation of individual contributions to the project and report.

Team Member	Role	Tasks Completed
Ben Snowden	Shaper	-Completed Report Executive Summary, Sections I-III, Sections V-VII, -Contributed to Adding the Appendices -Prototyped “Provide Alert” functionality
Austin Campbell	Team Worker	-Prototyped “Accept Programming” functionality
Everett Craig	Implementer	-Completed Majority of the Appendices
Sam Waldron	Coordinator	-Formatted and created final presentation slides
Sam Hall	Wild Card	-Prototyped “Ensure Portability” functionality
James Nguyen	Specialist	-Completed Section IV – Cost of Final Build